TCG-Net — Results Summary

Two-branch cooperative game-theoretic fusion of resting-state functional connectivity and plasma proteomics · UK Biobank

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All results below are held-out test-set numbers. Joint-model figures are aggregated over **11 independent training runs** differing only in random seed (five runs for the interpretability output in Section 1), each trained to convergence under its own early-stopping criterion. Participants used to evaluate the joint model are excluded from every pretraining and baseline fit, so no evaluation subject influences any model it is scored on.

1. Shapley contributions and model interpretability

For each participant the fusion module estimates the contribution of the connectome (φ_f), of plasma proteomics (φ_p) and of their interaction (φ_{int}), then converts them into the gate weights that drive latent fusion. These are the model's interpretability output. Distributions below pool five converged runs on the held-out fold.

Contribution	mean	sd	p5	median	p95
φ_f — connectome	−0.0552	0.2610	−0.5312	−0.0376	+0.3371
φ_p — proteomics	−0.0282	0.0882	−0.1669	−0.0302	+0.1141
φ_{int} — interaction	−0.0812	0.1652	−0.3388	−0.0858	+0.1903

The interaction term is a live measured quantity

For two players the Shapley efficiency axiom gives $\varphi_f + \varphi_p = \hat{y}_{fp} - \hat{y}_0$ identically, so defining the interaction as the residual of that identity returns exactly zero for every participant — measured here at 6.0×10^{-8}, zero to numerical precision. The model instead uses the two-way interaction index $I = \hat{y}_{fp} - \hat{y}_f - \hat{y}_p + \hat{y}_0$, which varies substantially between participants (sd 0.165, range −0.94 to +0.54). **Its spread exceeds that of the proteomic main effect** (sd 0.088): cross-modal interaction differs more between individuals than the proteomic contribution itself does.

Gate weights are not measures of predictive contribution

Branch	mean w	sd w	range	branch r
Connectome	0.335	0.047	0.169 – 0.518	+0.259
Proteomics	0.342	0.029	0.256 – 0.443	+0.100
Interaction	0.324	0.024	0.205 – 0.415	—

The proteomics branch carries the **highest** mean gate weight (0.342) while correlating with the target at less than half the connectome branch's rate (0.100 against 0.259). The weights are a softmax over the Shapley values, so a branch can hold a high weight while predicting poorly.

> A high w_p must not be reported as evidence of proteomic importance. It is a property of the gate, not of the branch's predictive skill.

All three weights sit near one third because the gate operated at temperature 1.0 throughout these runs; the sharpened setting was not active.

The subtyping score reduces to a single branch

Substituting the contribution definitions, the subtype score $D = \varphi_f - \varphi_p$ simplifies exactly to $\hat{y}_f - \hat{y}_p$ — confirmed numerically, with a maximum discrepancy of 6.0×10^{-8}. Because the proteomics branch output has a collapsed amplitude, D correlates with the connectome prediction at **0.998**, and its correlation with the target (0.254) is indistinguishable from the connectome branch's own (0.259).

Subtype	n	share	mean target (z)
Brain-dominant	247	33.1%	+0.207
Balanced	281	37.6%	+0.038
Blood-dominant	219	29.3%	−0.372

The three subtypes partition participants by their connectome prediction rather than by a brain-versus-blood mechanism, so the differences in cognitive score above are a selection effect. Rescaling the two branch outputs to comparable amplitude before differencing would make D a genuine two-modality contrast, and is the natural next step for this analysis.

The group sizes are also uninformative on their own: thresholding a standardised score at ±0.5 yields approximately 31 / 38 / 31 per cent for any normally distributed variable, so a near-equal three-way split is the expected outcome rather than evidence of structure.

2. Fluid intelligence — prediction results

The joint model is trained on 3,471 participants and evaluated on 747 held-out participants. Alongside the fused prediction it exposes three internal predictions: an fMRI-only head, a proteomics-only head, and a plain MLP on the concatenated embeddings. Scoring all four on the same subjects shows where the performance originates.

Output	R^2	R^2 calibrated	Pearson r
Fused prediction (full model)	**0.0725 ± 0.0056**	**0.0768**	**0.279**
fMRI branch only	0.0652 ± 0.0028	0.0665	0.260
Proteomics branch only	−0.0010 ± 0.0022	0.0285	0.172
Concatenation head (naive fusion)	0.0665 ± 0.0060	0.0706	0.267

The calibrated column matters. An affine rescaling fitted on the validation fold and applied to the test fold corrects a systematic amplitude error in the model outputs. It changes the reading of the proteomics branch entirely: a raw R^2 of approximately zero becomes 0.0285, essentially the theoretical maximum for a predictor correlating at r = 0.17. The proteomics branch is therefore **extracting all of the signal available to it** — its raw score of zero reflected a collapsed output scale, not an absence of information.

Comparisons between the model's own outputs

Contrast	Mean Δ R^2	Seeds in favour	t (df = 10)
Fused − fMRI branch	+0.0073	9 / 11	+3.82
Fused − fMRI branch (calibrated)	**+0.0103**	**11 / 11**	—
Concatenation head − fused	**−0.0060**	**1 / 11**	**−4.19**

The cooperative fusion output exceeds its own fMRI branch on 11 of 11 seeds once calibrated, and exceeds the naive concatenation head on 10 of 11. Within the architecture, the Shapley-guided gate outperforms both the single-modality path and the standard concatenation alternative.

Why the proteomic contribution is small

The proteomics-to-cognition association decomposes cleanly, and the loss is attributable rather than anomalous.

Setting	Pearson r	Explanation
Full proteomics cohort, baseline visit	0.345	reference
Imaging subcohort, baseline visit	0.254	cohort restriction
Imaging subcohort, imaging visit	0.171	~ 9-year blood-to-scan interval

The final step is fully accounted for by measurement attenuation: the test–retest correlation of the cognitive measure between the two visits is 0.636 on these participants, and 0.254 × 0.636 = 0.162 against 0.171 observed. A predictor correlating at r = 0.17 is capped at $R^2 = 0.029$ however it is modelled. **This is a property of the cohort design** — plasma was sampled roughly nine years before the scan — **and not of the model.**

3. Ablation and baseline testing — fluid intelligence

Every model below is evaluated on the same 747 held-out participants. Two families of linear baseline were run. The first is fitted on the joint training fold only (3,471 participants). The second is given the same advantage the deep encoders receive: fitted on the full single-modality cohort with all joint validation and test participants excluded, then adapted to the joint training fold by a two-parameter affine map. The second family is the fair comparison.

Model	Fitted on	R^2	Pearson r
Predict the training mean	—	−0.0007	—
Ridge, proteomics (2,920 features)	3,471	0.0242	0.165
Ridge, connectivity (2,145 edges)	3,471	0.0409	0.213
Ridge, concatenated features	3,471	0.0317	0.183
Ridge, proteomics — pretrained	11,528	0.0363	0.195
Ridge, connectivity — pretrained	30,511	0.0818	0.290
Late fusion of the two pretrained ridges	both	**0.1035**	**0.325**
TCG-Net, fused output	3,471	0.0725	0.279
TCG-Net, fused output (calibrated)	3,471	**0.0768**	—

What the baselines establish

(a) Multimodal integration genuinely helps. Late fusion of the two pretrained linear models reaches 0.1035 against 0.0818 for connectivity alone — a gain of +0.0217. Plasma proteomics does add information beyond the connectome for cognitive prediction.

(b) Feature-level concatenation is not the way to combine them. Fitted on the joint training fold, concatenating raw features (0.0317) performs **worse** than connectivity alone (0.0409). Naive early fusion destroys signal at this sample size, which motivates a learned fusion mechanism.

(c) The current network does not yet beat the strongest linear baseline. Late fusion of pretrained ridges reaches 0.1035 against 0.0768 for the calibrated fused output — a shortfall of 0.027 R^2, several times the seed-to-seed standard deviation of 0.006.

> The likely explanation is **representational rather than architectural**. The linear model uses all 2,145 connectivity edges with weights estimated from 30,511 participants, whereas the graph encoder compresses the connectome to 128 dimensions and its downstream heads are refitted on 3,471. The fusion mechanism itself performs well in every internal comparison (Section 1); it is the encoded representations that trail the raw features.

Ablations still outstanding. Single-modality, early-fusion and late-fusion comparisons are covered above. Not yet run: attention-based fusion, a learned non-Shapley modality gate, and the model without the interaction term, the squeeze-and-excitation block, or the graph encoder. The learned non-Shapley gate is the most informative of these, since it isolates the contribution of the game-theoretic weighting specifically.

4. Ablation and baseline testing — age

Age was assessed as a second target because both modalities carry substantial signal for it, which fluid intelligence does not permit.

Single-modality proteomics encoder

Model	R^2	MAE (years)	Pearson r
Predict the training mean	−0.0000	6.99	—
Ridge, all 2,920 proteins	**0.8690**	**2.35**	**0.932**
Ridge, 200 strongest proteins	0.7677	3.14	0.876
Proteomics encoder (Stage 1)	**0.868**	—	—

Evaluated on 7,904 held-out participants, **the proteomics encoder and ridge regression are indistinguishable** (0.868 against 0.8690). Age is linearly decodable from an Olink panel, so this figure reflects the information content of the assay rather than the encoder architecture. This was the specific comparison requested, and the answer is that the proteomics branch does not require a deep architecture.

Joint cohort — the multimodal baselines

Evaluated on 799 held-out participants from the joint cohort, using the same two baseline families as in Section 3.

Model	Fitted on	R^2	Pearson r
Predict the training mean	—	−0.0001	—
Ridge, connectivity	3,730	0.2469	0.498
Ridge, proteomics	3,730	0.7742	0.880
Ridge, concatenated features	3,730	0.7828	0.885
Ridge, connectivity — pretrained	33,081	0.3526	0.595
Ridge, proteomics — pretrained	35,754	0.8106	0.901
Late fusion of the two pretrained ridges	both	**0.8228**	**0.908**

The multimodal gain is positive for age. Late fusion improves on proteomics alone by +0.0122 R^2, and unlike fluid intelligence, concatenating raw features also helps rather than hurts (0.7828 against 0.7742). Both modalities carry real age information and are partially complementary — although the connectome's contribution is largely redundant with the proteome's, since only about 3.5% of its standalone signal survives as independent.

Pretraining is worth a great deal on the imaging side specifically: connectivity ridge improves from 0.2469 to 0.3526 when fitted on 33,081 rather than 3,730 participants. This is why the pretrained baseline family was constructed — comparisons against joint-fold-only baselines systematically overstate any deep model's advantage.

Status of the joint age model

The two-branch model has not yet been run on age. It requires a connectome encoder pretrained against age, which is pending on the imaging side. The tables above therefore establish the threshold rather than a comparison:

> The joint age model must exceed $R^2 = 0.8228$ (r = 0.908) to demonstrate value over a purely linear approach.

Summary

The pipeline now produces properly held-out, multi-seed, leakage-controlled results on both targets. Three findings stand out.

1. The cooperative fusion mechanism outperforms both single-modality prediction and naive concatenation within the architecture, consistently across seeds. The interaction term is a live, individually varying quantity whose spread exceeds that of the proteomic main effect.
2. Multimodal integration is genuinely informative for both targets. The proteomic contribution to cognition is small for a quantifiable reason — a nine-year interval between blood sampling and scanning that attenuates the association to a ceiling of $R^2 = 0.029$.
3. Neither deep encoder currently outperforms a regularised linear model given the same training data, which locates the remaining work in the learned representations rather than in the fusion mechanism.
4. Two interpretability quantities need restating before publication: the gate weights do not measure predictive contribution, and the subtyping score currently reduces to the connectome prediction alone.

Immediate priorities. Complete the remaining ablations, particularly the learned non-Shapley gate; obtain an age-trained connectome encoder so the joint age model can be evaluated against the 0.8228 threshold; and test the branch comparisons at the participant level by bootstrap, since the figures reported here quantify variation across training runs rather than across participants.